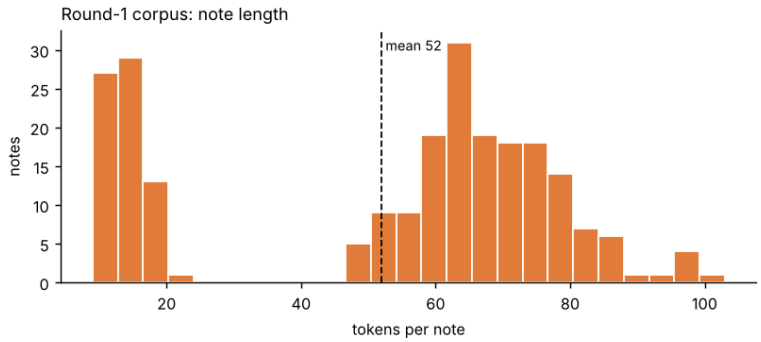
MedBhasha — EDA Summary (corpus v2, for Round 2)

Team JUSL Health Home · synthetic notes, no clinician review

*This summary describes the **final** validated corpus `data/round1/annotated_corpus.csv` (the Round-1 notes n001–n040 with 8 documented corrections + 192 new notes). Numbers are generated by code from the file.*

1. Corpus size

- Notes: **232** (Round 1 submission: 40) · tokens: **12060** (Round 1: 612; organisers' recommendation ≈ 10 000)
- Tokens per note: mean 52, median 62, min 9, max 103
- Entity spans: **2899** (12.5 per note on average; min 2, max 25); notes without any entity: 0
- Domains (project label, not an official field): 23 — largest: Respiratory 30, Gastroenterology 27, Cardiology 26, Dermatology 23

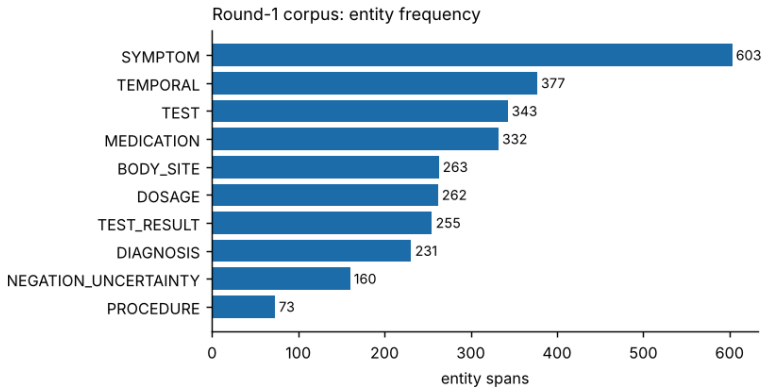


Note length

2. Entity frequency distribution

entity type	spans	distinct forms	notes containing	share %
SYMPTOM	603	221	206	20.8
TEMPORAL	377	114	193	13.0
TEST	343	144	162	11.8
MEDICATION	332	184	173	11.5
BODY_SITE	263	147	169	9.1
DOSAGE	262	135	153	9.0
TEST_RESULT	255	188	151	8.8
DIAGNOSIS	231	179	189	8.0

entity type	spans	distinct forms	notes containing	share %
NEGATION_UNCERTAINTY	160	20	123	5.5
PROCEDURE	73	50	59	2.5



Entity frequency

Every one of the 10 types now has at least 73 spans (Round 1: most types < 30). PROCEDURE is the rarest type; in outpatient notes most clinical acts are tests or prescriptions.

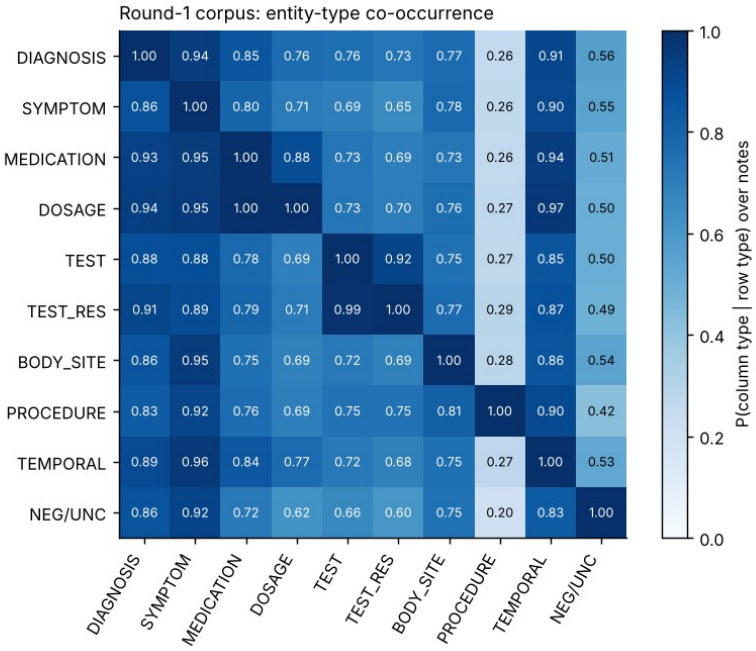
3. Domain distribution

domain	notes	share %
Respiratory	30	12.9
Gastroenterology	27	11.6
Cardiology	26	11.2
Dermatology	23	9.9
General Medicine	21	9.1
Infectious	19	8.2
Endocrine	14	6.0
Neurology	8	3.4
Psychiatry	8	3.4
Orthopaedics	8	3.4
Paediatrics	8	3.4
Renal	7	3.0
Obstetrics	6	2.6
Eye	6	2.6
Gynaecology	5	2.2
Hematology	3	1.3
ENT	3	1.3
Surgery	3	1.3

domain	notes	share %
Urology	2	0.9
Emergency	2	0.9
Rheumatology	1	0.4
Hepatology	1	0.4
Dental	1	0.4

4. Co-occurrence and relationships between entity types

Within-note co-occurrence (diagonal = notes containing the type). Row-normalised heat-map below: *P(column type present | row type present)*.



Co-occurrence

Most frequent type pairs:

type A	type B	notes containing both
SYMPTOM	TEMPORAL	186
DIAGNOSIS	SYMPTOM	178
DIAGNOSIS	TEMPORAL	172
MEDICATION	SYMPTOM	164
MEDICATION	TEMPORAL	163
DIAGNOSIS	MEDICATION	161
BODY_SITE	SYMPTOM	161

type A	type B	notes containing both
DOSAGE	MEDICATION	153
TEST	TEST_RESULT	149
DOSAGE	TEMPORAL	148

Reading: P(DOSAGE | MEDICATION) = 0.88, P(TEST_RESULT | TEST) = 0.92, P(TEMPORAL | SYMPTOM) = 0.90, P(NEGATION_UNCERTAINTY | DIAGNOSIS) = 0.56. These are the patterns a prescription/exam template produces; with 232 notes they are descriptive, not clinical findings.

Diagnosis → medication pairs seen at least twice, ranked by lift (lift > 1 = more often together than by chance):

diagnosis	medication	co_notes	n_a	n_b	lift	pmi
স্থিতিশীল অ্যানজাইনা	সরবিট্রেট	2	2	2	100.500	6.650
হাইপোথাইরয়েডিজম	লিভোথাইরক্সিন	2	2	2	100.500	6.650
হারপিস জোস্টার	অ্যাসাইক্লোভির	2	2	2	100.500	6.650
তীব্র আর্টিকারিয়া	ক্যালামাইন লোশন	2	2	3	67.000	6.070
দাদ	টারবিনাফিন ক্রিম	2	3	2	67.000	6.070
মাইগ্রেন	সুমাট্রিপ্টান	2	3	2	67.000	6.070
ইসকিমিক স্ট্রোক	অ্যাটরভাস্টাটিন	2	2	4	50.250	5.650
কিডনিতে পাথর	টামসুলোসিন	2	2	4	50.250	5.650

5. Data-quality checks

- empty tokens: 0 · invalid tags: 0 · official `validate.py`: **PASS, 0 errors, 0 warnings**
- exact duplicate notes: 0 · max character-4-gram Jaccard between any two notes: **0.45** (threshold 0.6)
- max character-5-gram Jaccard between any of our notes and any organiser-released test note: **0.28** (no copying)
- every entity type has ≥ 50 spans; the corpus contains short informal notes, structured outpatient notes, shorthand (`C/O … Dx … Rx`) notes and admission/discharge summaries.
- consistency audit: same-surface/different-type conflicts were inspected; the one systematic problem (organ + symptom written both as one SYMPTOM and as BODY_SITE + SYMPTOM) was corrected — see `docs/ANNOTATION_CONVENTIONS.md` rule C7.

6. Annotation edge cases (for clinician review)

topic	decision	where
organ + symptom (বুকে ব্যথা)	split: BODY_SITE + SYMPTOM; idioms such as	C7

topic	decision	where
	মাথা ঘোরা stay one SYMPTOM	
units in results (১১০/৭০ মি.মি. পারদ)	unit is inside TEST_RESULT	C6
imaging name with organ (বুকের এক্স-রে)	বুকের = BODY_SITE, এক্স-রে = TEST (organisers' worked example)	C5
endoscopy / dressing / nebulisation	PROCEDURE; imaging and lab → TEST	C10
negation	marker only (নেই, না, সন্দেহ)	C11
history diagnoses	DIAGNOSIS (no separate history label in the schema)	C9
সকালে, রাতে	TEMPORAL for symptoms, DOSAGE for when to take a drug	C2/C3
hyphen suffix tokens (ইসিজি-তে)	whole token tagged	C13

7. Limitations

- Notes are **synthetic** and written by a single author; the corpus measures consistency with one author's patterns, not real clinical language.
- **No clinician review**, no second annotator, no inter-annotator agreement.
- Domain balance is uneven; some specialties have only 2–6 notes.
- Whitespace tokenisation keeps Bengali case endings attached (ডায়াবেটিসের), so span boundaries include them (same as the organisers' worked example).